

Agentic AI for Every Classroom

A Free 6-Week Curriculum for High School Students

This kit contains everything a teacher needs to run a complete 6-week unit on agentic AI — no prior AI expertise required. Students will progress from "what is an AI agent?" to building and demoing their own working agent using free, browser-based tools.

What's in this kit	Format
6 weekly lesson plans (objectives, outline, discussion questions)	PDF
6 student activity worksheets	PDF
Assessment rubrics for each week	PDF
Standards alignment (CSTA, AI4K12, ISTE)	PDF
No-code tools setup guide	PDF
Editable slide decks	Google Slides (linked)

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ABOUT THIS CURRICULUM

What Is AgentEdu?

AgentEdu is a free, open-curriculum initiative making agentic AI education accessible to every high school student. This course was designed by an active AI Platform Architect who builds production agentic AI systems at enterprise scale — so the content is grounded in real technology, not textbook theory.

Why Agentic AI Specifically?

- 56%** higher wages for workers with AI skills vs those without (PwC 2025 Global AI Jobs Barometer)
- 80%** of K-12 students say teachers did NOT teach them how AI works (RAND 2025)
- 90%+** of global enterprises face critical AI skills shortages by 2026 (IDC)
- 143%** year-over-year growth in demand for AI engineers (DemandSage 2025)

How to Use This Curriculum

- **Full course:** Run all 6 weeks sequentially in an AP CS, intro to AI, or tech elective class.
- **Standalone module:** Each week works independently. Week 2 (prompting) and Week 5 (build session) are especially popular as single-day workshops.
- **After-school club:** Works well in a 60-minute/week AI club format.
- **Guest workshop:** Week 1 or Week 5 can be delivered as a 90-minute industry guest visit.

No AI expertise required.

Every lesson plan contains the background knowledge you need to facilitate. If a student asks something you don't know, the answer is usually: "Let's ask the AI and see what happens." That's a valid teaching moment.

WEEK 1

What Is an AI Agent?

From chatbots to agents — understanding autonomy and the reasoning loop.

Learning Objectives

- Define an AI agent and distinguish it from a simple AI tool or chatbot
- Identify and explain the three stages of the agent loop: Think → Act → Observe
- Give 3 real-world examples of AI agents and map their decision process

Key Concepts

AI Agent

A system that perceives its environment, plans, takes actions, and adapts based on feedback

Agentic AI

AI that has autonomy — it takes sequences of actions toward a goal without constant human direction

Reasoning Loop

Think (plan) → Act (do something) → Observe (see what happened) → repeat

LLM

Large Language Model — the "brain" powering modern AI agents

Hallucination

When an AI generates plausible-sounding but false information

Lesson Outline (75 min)

Time	Phase	What Happens
0:00–0:10	Hook	Demo a real AI agent doing something surprising. Ask: "What did it just do that a basic chatbot couldn't?"
0:10–0:25	Lecture	The Think → Act → Observe loop. Compare: calculator (not agentic) vs. AI research assistant (agentic).
0:25–0:40	Group work	Show 3 real agents. Small groups identify their tools, decisions, and failure modes.
0:40–0:65	Activity	The Agent Hunt worksheet (see Week 1 Worksheet PDF).
0:65–0:75	Share-out	Groups present their agent profiles. Class discussion.

Discussion Questions

1. What's the difference between asking an AI a question and giving it a task to complete on your behalf?
2. Can you think of a situation where an AI agent acting autonomously could cause harm?
3. What would have to be true for you to trust an AI agent to act on your behalf without asking?

◆ This Week's Activity

The Agent Hunt: Students choose a technology they use and fill in an Agent Profile — identifying its goal, Think/Act/Observe loop, autonomy level, and one potential failure mode.

Tools: Claude.ai or ChatGPT (teacher demo only)

WEEK 2

The Brain Behind the Agent

How LLMs work, why they hallucinate, and how to prompt effectively.

Learning Objectives

- Explain how a Large Language Model works at a conceptual level (next-token prediction)
- Write effective prompts using role assignment, step-by-step instruction, and few-shot examples
- Identify when AI output is a hallucination and explain the structural reason it happens

Key Concepts

Next-token prediction

LLMs generate text by predicting what word comes next — extremely sophisticated autocomplete

Context window

How much text the model can "see" at once — its working memory

Hallucination

Structurally caused by prediction over real retrieval — the model generates what sounds right, not what's true

Chain-of-thought

Prompting the model to reason step-by-step before answering — dramatically improves accuracy

Few-shot prompting

Including examples in your prompt to guide the model's output format or reasoning style

Lesson Outline (75 min)

Time	Phase	What Happens
0:00–0:10	Hook	Ask the AI a question it gets confidently wrong. Discuss: why did it fail?
0:10–0:30	Lecture	LLM mechanics — next-token prediction explained simply. Why hallucination is structural, not a bug.
0:30–0:50	Demo	Live: show the difference between a vague prompt and a well-engineered one. Four techniques: role, steps, format, examples.
0:50–0:70	Activity	Prompt Engineering Challenge — three rounds (explainer, debugger, planner).
0:70–0:75	Debrief	What made the winning prompts work?

Discussion Questions

1. If an LLM is "predicting the next word," why does it seem to understand what we mean?
2. How is hallucination different from a human making a mistake? Is it more dangerous, or less?
3. If you were building an AI agent for medical advice, what safeguards would you add?

◆ This Week's Activity

Prompt Engineering Challenge: Three rounds. Students compete to get the best output using only prompt engineering — explaining quantum computing to a child, debugging code, and creating a study plan.

Tools: Claude.ai or ChatGPT — free accounts

WEEK 3

Tools & Memory

Why bare LLMs are limited — and how tools and memory transform them.

Learning Objectives

- Explain why an LLM alone is limited and what tools add to agent capability
- Distinguish between short-term (in-context) and long-term (persistent) memory
- Describe how RAG (Retrieval-Augmented Generation) works using a real-world analogy

Key Concepts

Tool calling

The ability to invoke external functions (search, calculator, code, database) and use the results

Context window

The amount of text an LLM can "see" at once — its short-term working memory

RAG

Retrieval-Augmented Generation — fetching relevant documents and giving them to the LLM before it answers

Vector database

Stores information as mathematical embeddings, enabling similarity-based retrieval

Persistent memory

Information stored outside the LLM and retrieved across multiple conversations

Lesson Outline (75 min)

Time	Phase	What Happens
0:00–0:10	Hook	Ask an AI about yesterday's news. Watch it fail. Then show it with web search enabled. What changed?
0:10–0:30	Lecture	LLM limitations: no real-time data, no compute, no persistent memory. How each tool solves one problem.
0:30–0:45	RAG concept	The open-book test analogy: "RAG is giving the AI the right textbook pages before it answers."
0:45–0:70	Activity	Build a Memory Bot — test what an AI remembers across conversations.
0:70–0:75	Debrief	What did it remember? What would make this useful in a real product?

Discussion Questions

1. What kinds of tasks would you NOT trust an AI agent without tools to do?
2. If an AI can search the internet in real time, does that fully solve hallucination? Why or why not?
3. What personal information would you be comfortable storing in an AI's long-term memory about you?

◆ This Week's Activity

Memory Bot Build: Students use a free platform (Claude.ai or ChatGPT with memory enabled), give the AI specific facts to remember, end the session, start a new one, and test what persisted.

Tools: Claude.ai (memory enabled) or ChatGPT with memory

WEEK 4

Multi-Agent Systems

When one agent isn't enough — orchestration, handoffs, and failure.

Learning Objectives

- Explain why multi-agent systems exist and when a single agent is insufficient
- Distinguish between an orchestrator agent and a specialist worker agent
- Design a simple multi-agent pipeline with handoffs for a specific use case

Key Concepts

Multi-agent system

A network of AI agents collaborating to complete complex tasks no single agent could handle

Orchestrator

The "manager" agent — breaks down tasks and assigns subtasks to specialist agents

Worker/specialist

An agent with a specific skill: coding, writing, research, fact-checking, etc.

Handoff

Passing information or control from one agent to another in the pipeline

Human-in-the-loop

A design pattern where humans review or approve agent actions before execution

Lesson Outline (75 min)

Time	Phase	What Happens
0:00–0:10	Hook	"Why do companies have departments instead of one all-knowing employee?" Draw the parallel to AI teams.
0:10–0:30	Lecture	Real enterprise examples. Show: Orchestrator → Research → Summarize → Fact-check → Deliver.
0:30–0:50	Discussion	What goes wrong when agents hand off? Error compounding, trust, and the case for human checkpoints.
0:50–0:70	Activity	Design Your Agent Team — map a multi-agent system for a real use case.
0:70–0:75	Gallery walk	Groups share their designs. Class votes on "most likely to actually work."

Discussion Questions

1. If you had a team of AI agents handling your homework, what would each specialist do?
2. How do you decide which agent to trust if two agents disagree?
3. At what point does a multi-agent system become too autonomous to be safe?

◆ This Week's Activity

Design Your Agent Team: In pairs, students pick a scenario (homework helper, travel planner, sports analyst) and map the full pipeline — orchestrator, workers, handoffs, and where a human should stay in the loop.

Tools: Paper + pencil (no tech required)

WEEK 5

Build Your First Agent

Hands-on project week — build, test, break, iterate, and demo.

Learning Objectives

- Use a no-code agent builder to create a functional AI agent from scratch
- Test their agent, deliberately identify failure modes, and iterate
- Present and demo their agent to peers with a structured explanation

Key Concepts

System prompt

The background instructions that define an agent's personality, rules, and goals

Iterative testing

Systematically trying to break your agent to find weaknesses before deployment

Agent robustness

How well an agent handles unexpected inputs, edge cases, and adversarial prompts

Demo readiness

Can you explain what your agent does, how it works, and what it can't do yet?

Lesson Outline (75 min)

Time	Phase	What Happens
0:00–0:10	Setup	Introduce the three build challenges. Tool walkthrough (5 min).
0:10–0:20	Demo	Teacher builds a simple research agent live — full transparency.
0:20–0:60	Build session	Students build their chosen agent. Circulate and coach.
0:60–0:75	Demo day	Each student/pair demos (2–3 min). Peer feedback.

Discussion Questions

1. What was the hardest part of building your agent — the idea, the instructions, or the testing?
2. What's one thing your agent does surprisingly well? One thing it still can't do?
3. If you were deploying this for real users, what would you fix first?

◆ This Week's Activity

Build Challenge: Choose Option A (Research Agent), B (Study Buddy), or C (Creative Agent). Build, test using the "break it on purpose" technique, complete the build worksheet, and present a 2-3 minute demo.

Tools: Claude.ai · ChatGPT · Zapier AI Agents · Make.com (all free)

WEEK 6

AI Safety, Ethics & Your Future

What goes wrong, who's responsible, and where careers in AI lead.

Learning Objectives

- Identify at least 4 things that can go wrong with AI agents in production
- Articulate the ethical responsibilities of AI builders using real examples
- Describe 3 career paths in AI and the skills each requires

Key Concepts

Hallucination

AI generating false information with high confidence — structural, not a bug

Bias

Systematic errors in AI output reflecting unfairness in training data or design

Prompt injection

An attack where malicious input tricks an agent into ignoring its instructions

Alignment

The challenge of making AI systems pursue the goals their builders actually intend

Responsible AI

Principles for ethical AI: fairness, transparency, accountability, privacy, safety

Lesson Outline (75 min)

Time	Phase	What Happens
0:00–0:15	Review	Revisit Week 5 failure modes. Categorize them on the board: misinformation / bias / privacy / autonomy.
0:15–0:35	Lecture	The 4 risk categories. Real examples from deployed systems.
0:35–0:55	Debate	The hospital medication agent scenario — three groups, three positions.
0:55–0:70	Careers	AI Engineer, ML Engineer, AI Safety Researcher, AI Ethics Officer — skills and paths.
0:70–0:75	Reflection	Final reflection worksheet.

Discussion Questions

1. Is there a type of decision that AI agents should never be allowed to make autonomously?
2. Who should be held responsible when an AI agent causes harm — the builder, the company, or the user?
3. What skills do you already have that would be valuable in an AI career?

◆ This Week's Activity

The Hospital Debate: Three groups argue (A) deploy the autonomous medication agent, (B) never deploy it, or (C) deploy it with specific constraints. Each group presents. Class votes after all three arguments.

Tools: No tech required

STANDARDS & ALIGNMENT

Curriculum Standards

CSTA K-12 CS Standards

- 3A-AP-13 · Create prototypes that use algorithms to solve computational problems
- 3B-AP-14 · Evaluate the correctness of a program by constructing tests
- 3A-IC-24 · Evaluate the ways computing impacts personal, ethical, social, economic practices
- 3B-IC-25 · Evaluate computational artifacts to maximize benefits and minimize harmful effects

AI4K12 Big Ideas

- Big Idea 1 · Computers perceive the world using sensors
- Big Idea 3 · Computers can learn from data
- Big Idea 4 · Intelligent agents require many kinds of knowledge to interact naturally with humans

ISTE Student Standards

- 1c · Knowledge constructor
- 1d · Innovative designer
- 1e · Computational thinker

REQUIRED TOOLS

Free Tools Setup

Tool	Used In	URL	Notes
Claude.ai	Weeks 2, 3, 5	claude.ai	Free tier sufficient; test from school network first
ChatGPT	Weeks 2, 3, 5	chatgpt.com	Enable memory in settings for Week 3 activity
Zapier AI Agents	Week 5	zapier.com/ai	Free plan works; create account before class
Make.com	Week 5 (opt.)	make.com	Optional for advanced students; free tier
Google Slides	All weeks	slides.google.com	Slide decks linked from agentedu.org/resources